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VYSOKÉ UČENÍ TECHNICKÉ V BRNĚ

FACULTY OF INFORMATION TECHNOLOGY
FAKULTA INFORMAČNÍCH TECHNOLOGIÍ

DEPARTMENT OF COMPUTER GRAPHICS AND MULTIMEDIA
ÚSTAV POČÍTAČOVÉ GRAFIKY A MULTIMÉDIÍ

ADVANCED INJURY SIMULATION OF GAME CHARACTERS

POKROČILÁ SIMULACE ZRANĚNÍ HERNÍCH POSTAV

BACHELOR'S THESIS
BAKALÁŘSKÁ PRÁCE

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BRNO UNIVERSITY OF TECHNOLOGY
FACULTY OF INFORMATION TECHNOLOGY

Bachelor's Thesis Assignment

172352

Institut:	Department of Computer Graphics and Multimedia (DCGM)
Student:	**Helešic Adam**
Programme:	Information Technology
Title:	**Advanced Injury Simulation of Game Characters**
Category:	Computer Graphics
Academic year:	2025/26

Assignment:

1. Familiarize yourself with game development in the Godot engine.
2. Study the issue of game character animation and damage during combat.
3. Research existing and possible solutions to the given problem.
4. Design a user interface and the final functionality of an application that demonstrates advanced character reactions to injuries.
5. Implement the application and demonstrate its use with different types of data.
6. Perform user or performance testing of the results.
7. Document and publish the final application.
8. Create a video representing the results of your work.

Literature:

- Adams, Ernest, and Joris Dormans. Game mechanics: advanced game design. New Riders, 2012. ISBN 0321820274, 9780321820273.
- Geijtenbeek, T., Vasilescu, D., Egges, A. Injury Assessment for Physics-Based Characters. In: Allbeck, J.M., Faloutsos, P. (eds) Motion in Games. MIG 2011. Lecture Notes in Computer Science, vol 7060. Springer, Berlin, Heidelberg. 2011. DOI: 10.1007/978-3-642-25090-3_7.
- Francis Laclé, Nicolas Pronost. Real-Time Musculoskeletal Model for Injury Simulation on 3D Human Characters. Motion In Games Conference (MIG), Nov 2012, Rennes, France.
- Aristidou, A., Lasenby, J., Chrysanthou, Y. and Shamir, A. Inverse Kinematics Techniques in Computer Graphics: A Survey. Computer Graphics Forum, 37: 35-58. 2018. DOI: 10.1111/cgf.13310.

Requirements for the semestral defence:
Items 1-4, experiments leading to item 5.

Detailed formal requirements can be found at https://www.fit.vut.cz/study/theses/

Supervisor:	**Chlubna Tomáš, Ing., Ph.D.**
Head of Department:	Černocký Jan, prof. Dr. Ing.
Beginning of work:	1.11.2025
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Approval date:	7.11.2025

Abstract

Do tohoto odstavce bude zapsán výtah (abstrakt) práce v anglickém jazyce.

Abstrakt

Do tohoto odstavce bude zapsán výtah (abstrakt) práce v českém (slovenském) jazyce.

Keywords

Sem budou zapsána jednotlivá klíčová slova v anglickém jazyce, oddělená čárkami.

Klíčová slova

Sem budou zapsána jednotlivá klíčová slova v českém (slovenském) jazyce, oddělená čárkami.

Reference

HELEŠIC, Adam. *Advanced Injury Simulation of Game Characters.* Brno, 2026. Bachelor's thesis. Brno University of Technology, Faculty of Information Technology. Supervisor Ing. Tomáš Chlubna, Ph.D.

Advanced Injury Simulation of Game Characters

Declaration

Prohlašuji, že jsem tuto bakalářskou práci vypracoval samostatně pod vedením pana X... Další informace mi poskytli... Uvedl jsem všechny literární prameny, publikace a další zdroje, ze kterých jsem čerpal. Pro jazykovou revizi technické zprávy jsem použil(a) nástroj ChatGPT. Při tvorbě odevzdaného programového kódu jsem využíval nástroj Github Copilot. [konkretizujte a upravte podle skutečnosti].

.....
Adam Helešic
December 22, 2025

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Chapter 1

Introduction

@TODO: after finishing the thesis

úvod, asi trochu k design philosophy a cílům

The goal of this thesis is the design and implementation of a dynamic injury system of game characters in the Godot Engine. The term *character* here describes game characters in general, and is not intended to be limited to anthropomorphs.

This demo is not player-centric, but rather aims to implement a standalone injury system and its effects on the creature's behavior and animation. That is not to say player experience and game design is thrown out of the window, only that the player controller is not the focus of this demo.

The Godot Engine was chosen for this project because

Chapter 2 dives into

Chapter 2

Game Character Injury Systems and Animation (Theory)

Combat and injury have been a part of human history even since prehistoric times. Singular fights, epic battles, and strifeful wars riddle our existence in stories – mythological through modern, in our personal lives, or in large-scale historical real life events.

2.1 Game Design

The injury system has to be integrated into the game design and have an *actual effect on the gameplay.* Game Feel: good game mechanics quotes / possibly sth from Advanced Game Mechanics book

2.2 Systems of Locomotion

simulating movement, procedural movement and animation, possible techniques (like IK), adjusting based on centroid (e.g., different foot placement and leaning if a leg is gone)

The injury system has to be dynamic, adapting to whichever injuries are applied to the character's body. Hence procedural animation fits this goal of adaptibility well. Rather than traditional animation with preset model movements, procedural animation can account for and adapt to injuries on the run, and make for a more believable and immersive interaction with the injured character.

2.2.1 Inverse Kinematics

![TODO] Describe IK from survey_IK!

Inverse kinematics is „rotational and translational motion of points, bodies (objects) and systems of bodies (groups of objects) without consideration of what causes the motion or any reference to mass, force or torque“

It had originally been used in robotics for moving arms to targets, but has also expanded into animation in computer graphics. In the latter field, it is very useful for animating *articulated figures* (figures with a specific, solid form like humans, animals, and generally creatures with limbs, as opposed to blobs of jelly, for example). These models are often complicated, and require efficient calculation of complex and multi-link models that can have specified constraints (e.g., a maximum angle that a link can turn by).

[description of IK basic mechanisms and illustrative images - end effectors, end points and target points, joints and links] The multi-link models in inverse kinematics are usually calculated by points (target positions to reach and the model's joints, known as *end effectors*) and their potential constraints. These end effectors are specified by the model (the joints, the links between them, and the constraints limiting their movement) or by the animation system (the target points that a multi-link model, such as an arm, needs to reach to accomplish its task).

A model is a series of links connected by joints. Joints are the main component allowing movement – their positions determine the positions and orientations of the links. Links are the „connective tissue“ between joints and are defined by their length, shape, volume, and mass properties. A *posture* is the skeletal configuration of the figure and must most often satisfy constraints to adhere to realistic movement, such as the torque limitations of each joint or the links not intersecting with other models in their path. A skeletal configuration is usually separated into *chains*, standalone parts attached to a static point on the model (e.g., the limbs of animals and anthropomorphs). The chains usually work in a hierarchy, and the movement of one joint affects the movement of the ones before it.

[illustration image of chained movement effects]

Usage of Inverse Kinematics

Forward kinematics (traditional animation where known, predefined transformations are applied to the model) can fall short in situations where we might need to dynamically adjust the target locations, such as characters standing on slopes or an appandage reaching for a target that can be a variable distance or height away. Inverse kinematics is adaptive, the movement of the links (such as limbs) is dynamically calculated and placed, making for natural-looking movement. This quality is what makes it an ideal animation method for our dynamic limb-based injury system.

[image example(s) of IK use in video games where an end effector target - legs touching the actual ground, nathan drake touching walls, dynamic walking systems]

2.3 Simulating Harm and Injury

possible systems - HP-based; body-part based; idk

probably going with a joint HP pool where each body part is some fraction of that pool

the problem with critical injuries and instant death – should we aim to simulate injury realistically and allow instant death in cases of striking a critical body part like the head, or should we enforce a design decision and disallow instant death to preserve game balance?

2.4 Enemy AI

deciding on actions and movement; limiting options based on harm; procedural animation of actions

2.5 Case Studies

Nastudujte existující a možná řešení dané problematiky – jak to resi ostatni hry?

Horizon Zero Dawn/Forbidden West Monster Hunter Dead Space Dragon's Dogma The Surge Underrail Crackdown Vagrant Story Fallout Shadow Tower: Abyss Hybrid Heaven Wasteland Bushido Blade Binary Domain Phoenix Point Xenoblade Chronicles X Mechwarrior Assassin's Creed Valhalla

Chapter 3

Injury System Design

Navrhněte uživatelské rozhraní a výslednou funkcionalitu aplikace, která demonstruje pokročilé reakce herní postavy na zranění.

Design justified by design principles (books).

Godot plugin design and mockups

Chapter 4

Implementation and Demonstration in Various Test Cases/Situations

No need to reinvent the wheel, just make sure it works well. Using Godot 4's SkeletonIK3D and LookAtModifier3D.

Chapter 5

User/Performance testing of the resulting implemented design

Chapter 6

Documentation and Publication

+ video

Bibliography